

Higher insulin sensitivity in vegans is not associated with higher mitochondrial density.



BACKGROUND/OBJECTIVES: Vegans have a lower incidence of insulin resistance (IR)-associated diseases and a higher insulin sensitivity (IS) compared with omnivores. The aim of this study was to examine whether the higher IS in vegans relates to markers of mitochondrial biogenesis and to intramyocellular lipid (IMCL) content. SUBJECTS/METHODS: Eleven vegans and 10 matched (race, age, sex, body mass index, physical activity and energy intake) omnivorous controls were enrolled in a case-control study. Anthropometry, bioimpedance (BIA), ultrasound measurement of visceral and subcutaneous fat layer, parameters of glucose and lipid homeostasis, hyperinsulinemic euglycemic clamp and muscle biopsies were performed. Citrate synthase (CS) activity, mitochondrial DNA (mtDNA) and IMCL content were assessed in skeletal muscle samples. RESULTS: Both groups were comparable in anthropometric and BIA parameters, physical activity and protein-energy intake. Vegans had significantly higher glucose disposal (M-value, vegans 8.11 ± 1.51 vs controls 6.31 ± 1.57 mg/kg/min, 95% confidence interval: 0.402 to 3.212, $P=0.014$), slightly lower IMCL content (vegans 13.91 (7.8 to 44.0) vs controls 17.36 (12.4 to 78.5) mg/g of muscle, 95% confidence interval: -7.594 to 24.550, $P=0.193$) and slightly higher relative muscle mtDNA amount (vegans 1.36 ± 0.31 vs controls 1.13 ± 0.36 , 95% confidence interval: -0.078 to 0.537, $P=0.135$). No significant differences were found in CS activity (vegans 18.43 ± 5.05 vs controls 18.16 ± 5.41 $\mu\text{mol/g/min}$, 95% confidence interval: -4.503 to 5.050, $P=0.906$). CONCLUSIONS: Vegans have a higher IS, but comparable mitochondrial density and IMCL content with omnivores. This suggests that a decrease in whole-body glucose disposal may precede muscle lipid accumulation and mitochondrial dysfunction in IR development.